



COLLEGE CODE :9504

COLLEGE NAME:Dr.G.U.Pope college of engineering

DEPARTMENT :CSE

STUDENT NM-ID:85F79D4CB52BD52AA994E080EF3F2AE9

ROLL NO:950423104006

DATE:15/09/2025

Completed the project named as phase 1

NAME :Live Weather Dashboard

SUBMITTED BY,

NAME:Athis Kumar C

MOBILE NO:9488836278

Live Weather Dashboard

1. Problem Statement

Users often rely on multiple sources to check weather conditions (temperature, humidity, wind, forecasts). However, this information is often fragmented or not presented in a simple, real-time manner. A Live Weather Dashboard will provide a centralized, easy-to-use platform that fetches real-time weather data and displays it in a clean, interactive way for quick decision-making.

2. Users & Stakeholders

Primary Users

- General public (daily weather updates).
- Travelers (flight/train trip planning).
- Outdoor workers (farmers, construction workers, delivery staff).
- Event organizers (sports, concerts, outdoor gatherings).

Stakeholders

- Developers/Designers building the dashboard.
- Weather API providers (e.g., OpenWeatherMap, WeatherAPI).
- Businesses integrating weather insights (travel, logistics).

3. User Stories

- As a user, I want to view the current temperature and weather conditions so that I can plan my day.
- As a traveler, I want to check weather forecasts for upcoming days to schedule trips.
- As a farmer, I want to see rainfall and humidity updates to make agricultural decisions.
- As a business owner, I want to embed the weather dashboard in my website for customers.
- As a mobile user, I want a responsive and lightweight interface so I can quickly check weather on the go.

4. MVP Features

- Current location weather (temperature, condition, humidity, wind speed).
- Search weather by city/location.
- 5-day weather forecast.

- Responsive UI (desktop + mobile).
- Real-time data refresh via weather API.
- Error handling (e.g., 'city not found').
- (Future enhancements: weather alerts, radar maps, AQI monitoring, dark/light theme, localization.)

5. Wireframes / API Endpoint List

Wireframes (Conceptual)

- Dashboard Page:
- Header: Search bar + Current location button.
- Main Card: Current weather (temp, icon, condition).
- Details: Humidity, Wind Speed, Feels Like, Sunrise/Sunset.
- Forecast Section: 5-day cards with min/max temp + weather icons.

API Endpoints (Example: OpenWeatherMap API)

- Current Weather: GET `https://api.openweathermap.org/data/2.5/weather?q={city}&appid={API_KEY}&units={metric}`
- 5-Day Forecast: GET `https://api.openweathermap.org/data/2.5/forecast?q={city}&appid={API_KEY}&units={metric}`
- Geolocation-based Weather: GET `https://api.openweathermap.org/data/2.5/weather?lat={lat}&lon={lon}&appid={API_KEY}`

6. Acceptance Criteria

- User can search for a city and view current weather data.
- Weather data updates in real-time (or at least every API refresh cycle).
- User can view a 5-day forecast with daily temperature and condition.
- Dashboard is responsive across devices (mobile, tablet, desktop).
- If an invalid city is entered, an error message is shown.
- Data is fetched from a trusted weather API and displayed in user-friendly format.